Mini Project 1: Setup and Build Your Own Cybersecurity Lab

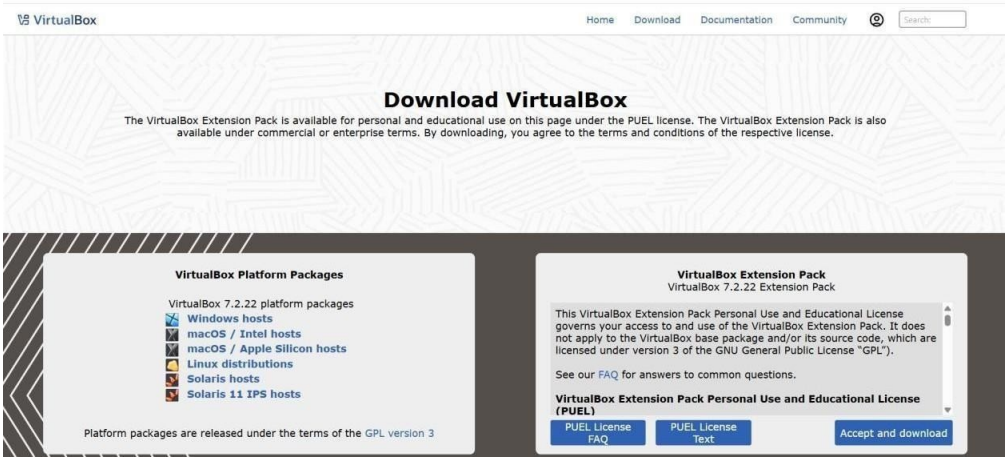
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Course / Group	Cyber Security / G17
Platform	SURE ProEd
Host OS	Windows 11
Date	22/10/2026

1. Aim

To prepare a safe, isolated practice environment by installing VirtualBox on Windows 11 and running Kali Linux inside a virtual machine.

Step 1: Download VirtualBox

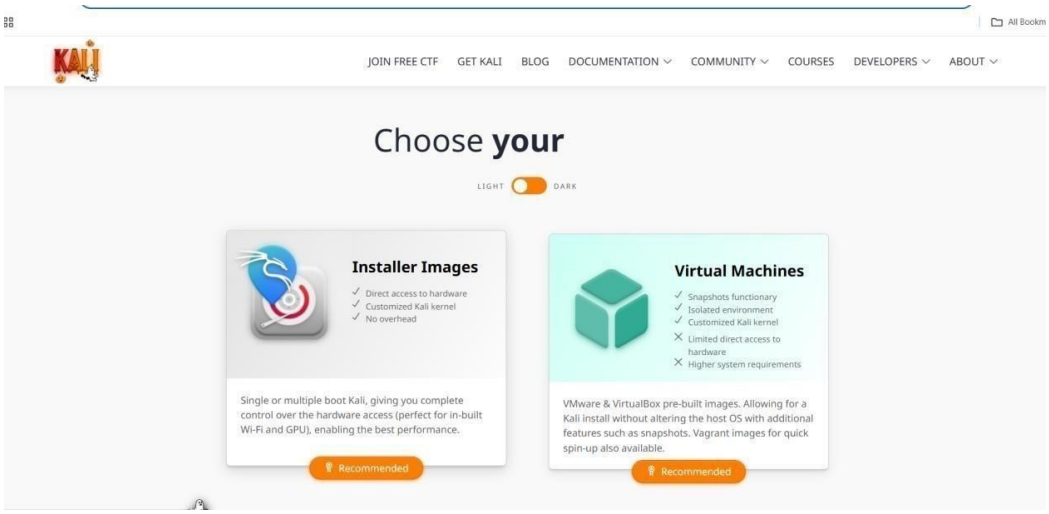
Open Google Chrome, search for VirtualBox and go to the official downloads page.
https://www.virtualbox.org/wiki/Downloads



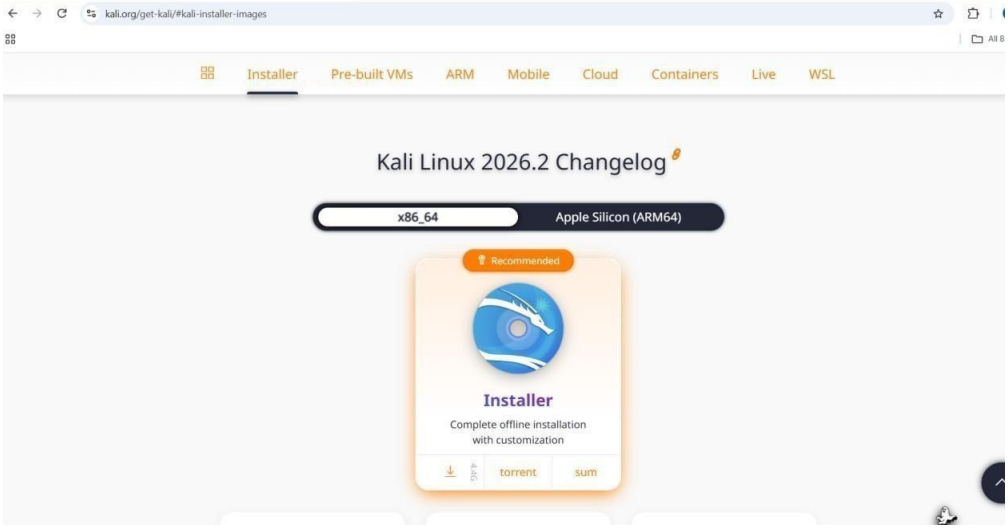
Choose the installer that suits the host operating system (Windows hosts). Also download the Extension Pack from the same page; the download starts as soon as the option is selected.

Step 2: Download Kali Linux

Search for Kali Linux and open the official download page.
https://www.kali.org/get-kali/#kali-platforms



The page lists several platforms. The recommended choice is the ready-made virtual machine image, but the installer images page below can be used as an alternative. https://www.kali.org/get-kali/#kali-installer-images

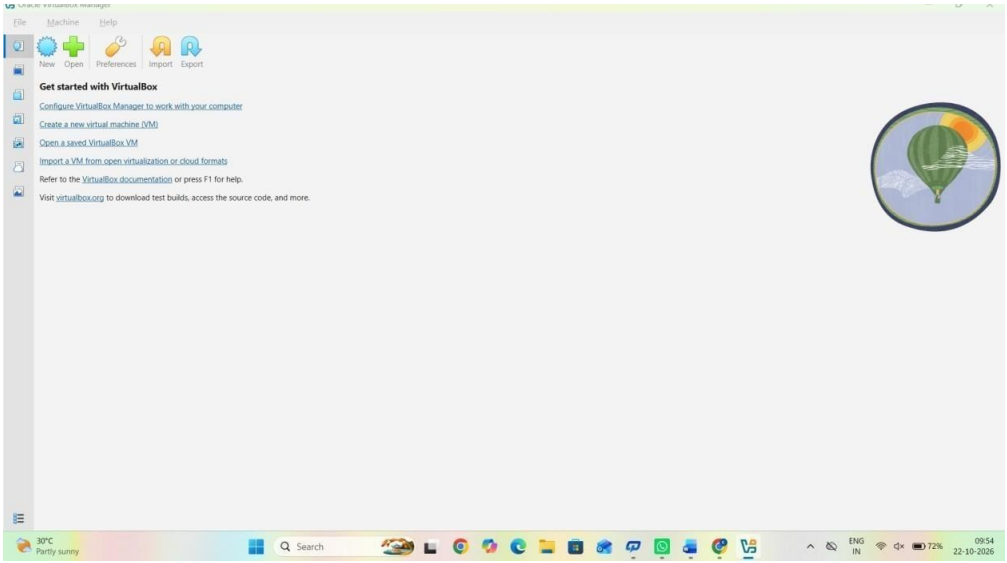


Select the required image. Once the download finishes, the file appears in the browser downloads list.

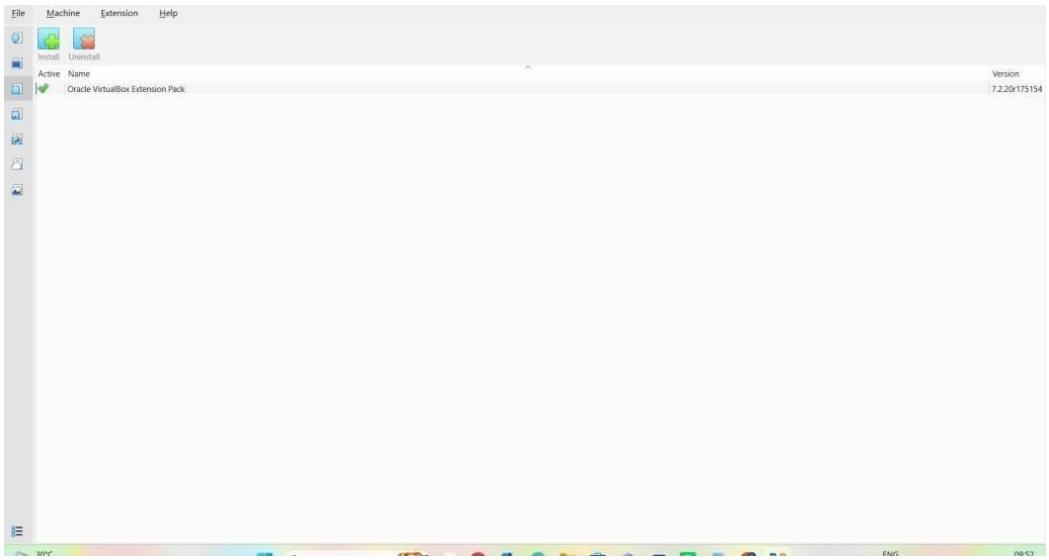


Step 3: Install VirtualBox and the Extension Pack

Open File Explorer, run the VirtualBox installer and keep clicking Next, accepting the defaults, until the installation completes.

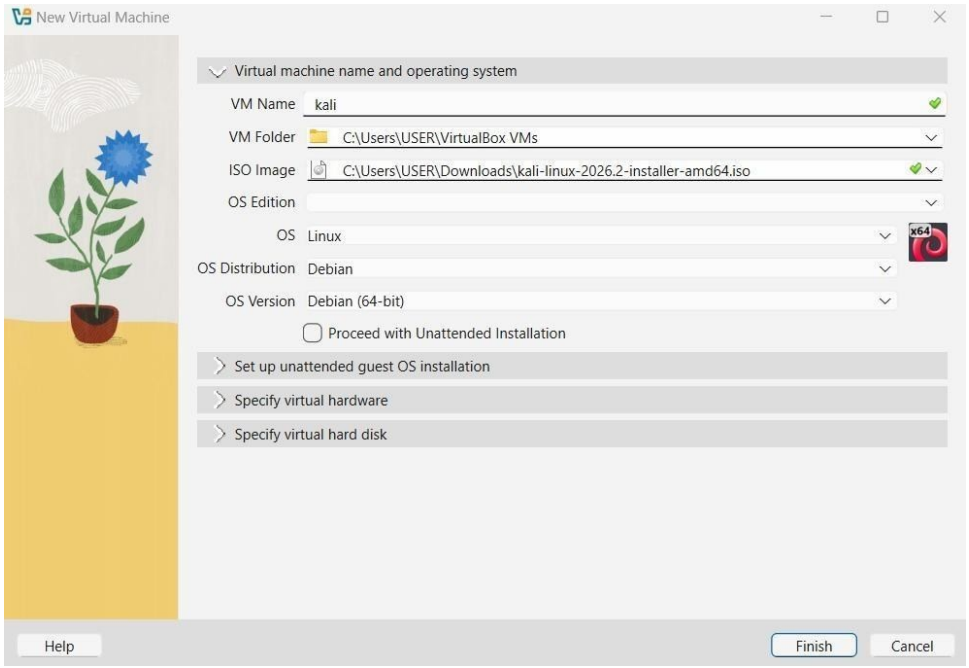


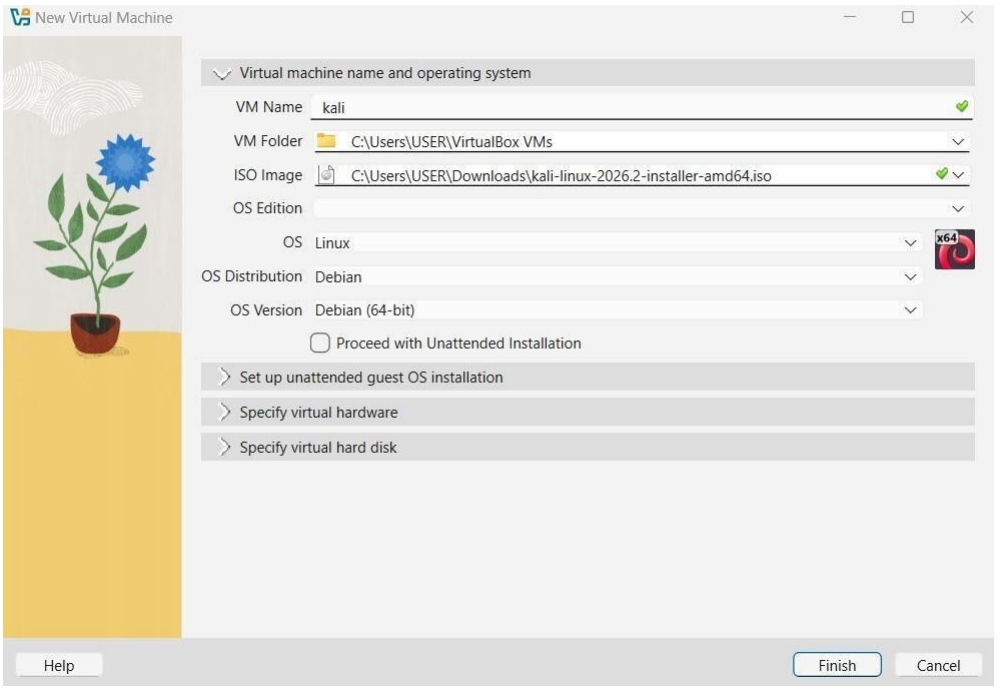
Then open the Extensions option in VirtualBox and select the downloaded Extension Pack file to install it. The window below confirms the extension was added.

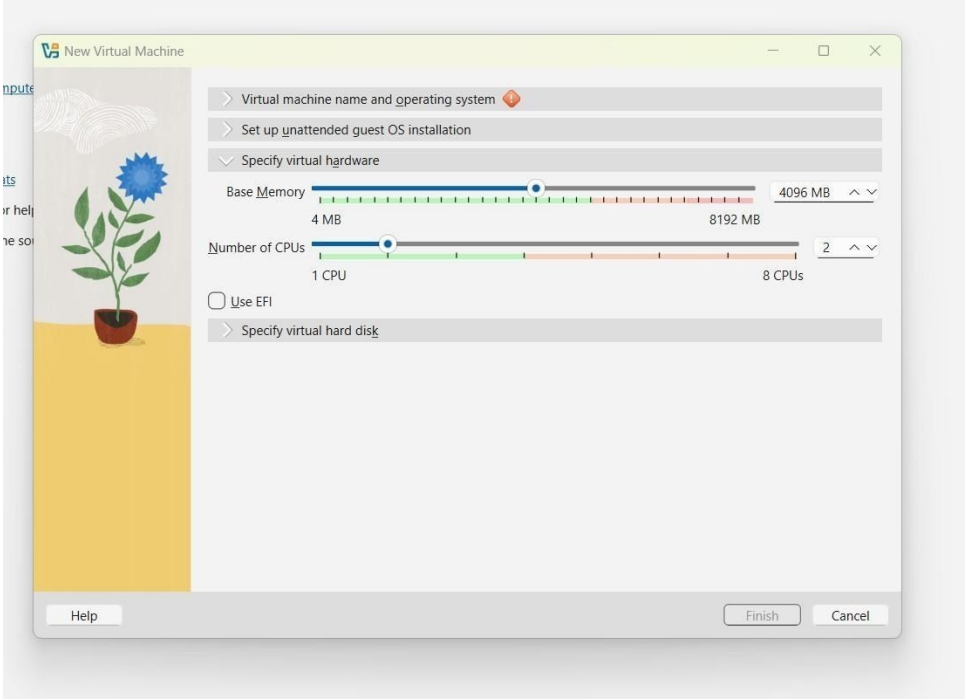


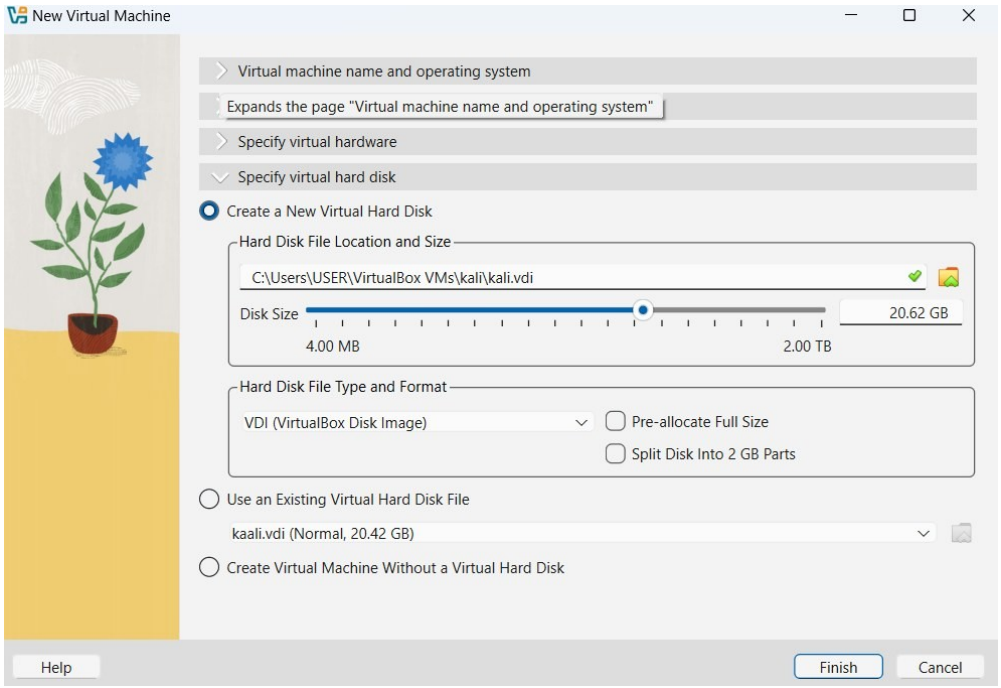
Step 4: Create the Virtual Machine

From the VirtualBox home screen click New to start the wizard. Enter the virtual machine name, select the Kali ISO image, and set the OS edition, distribution and version as shown in the screenshots.

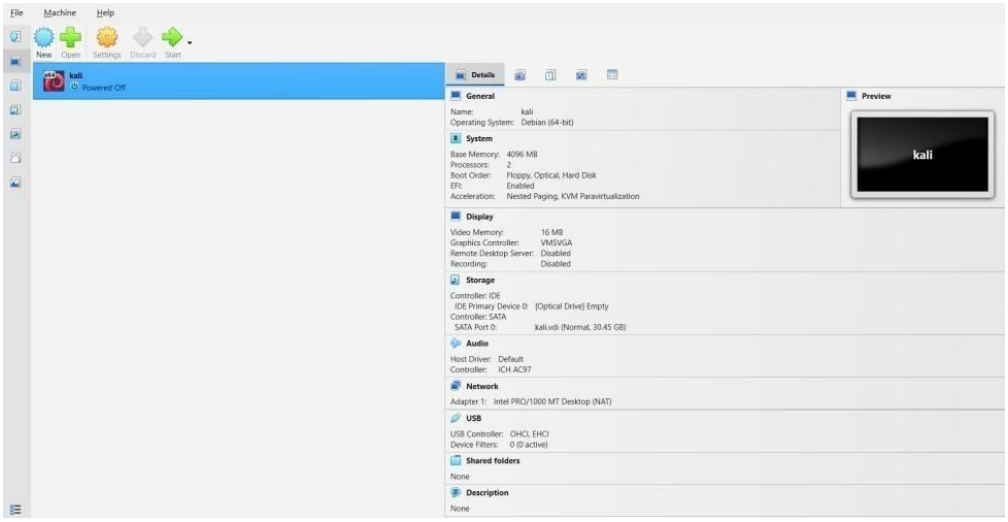



New Virtual Machine
Virtual machine name and operating system
VM Name
kali
VM Folder
C:\Users\USER\VirtualBox VMs
ISO Image
C:\Users\USER\Downloads\kali-linux-2026.2-installer-amd64.iso
OS Edition
OS
Linux
OS Distribution
Debian
OS Version
Debian (64-bit)
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Help
Finish
Cancel


New Virtual Machine
Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Base Memory
4096 MB
4 MB
8192 MB
Number of CPUs
2
1 CPU
8 CPUs
Use EFI
Specify virtual hard disk
Help
Finish
Cancel

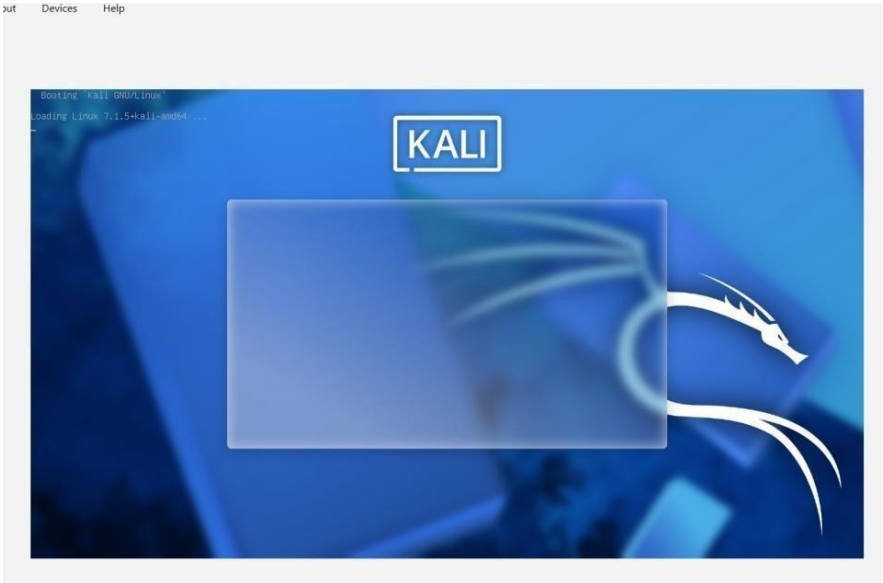


When the Finish button becomes active, click it. In the unattended installation settings enter a username and password of your choice (kept confidential here) and click Yes/Next until the setup completes.



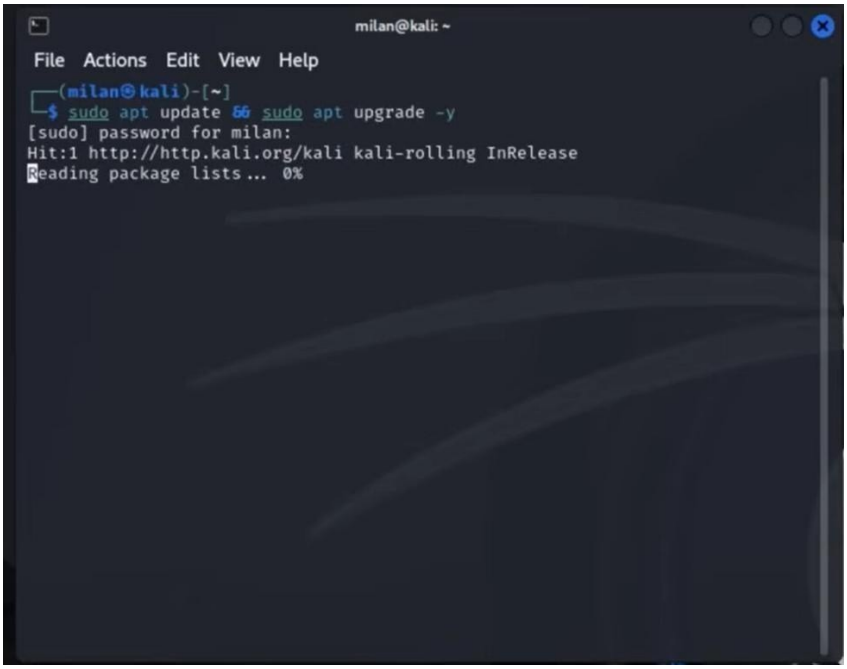
Step 5: Start Kali Linux and Update It

Click Start to power on the virtual machine and wait about five minutes for the first boot. Log in with the username and password created earlier.



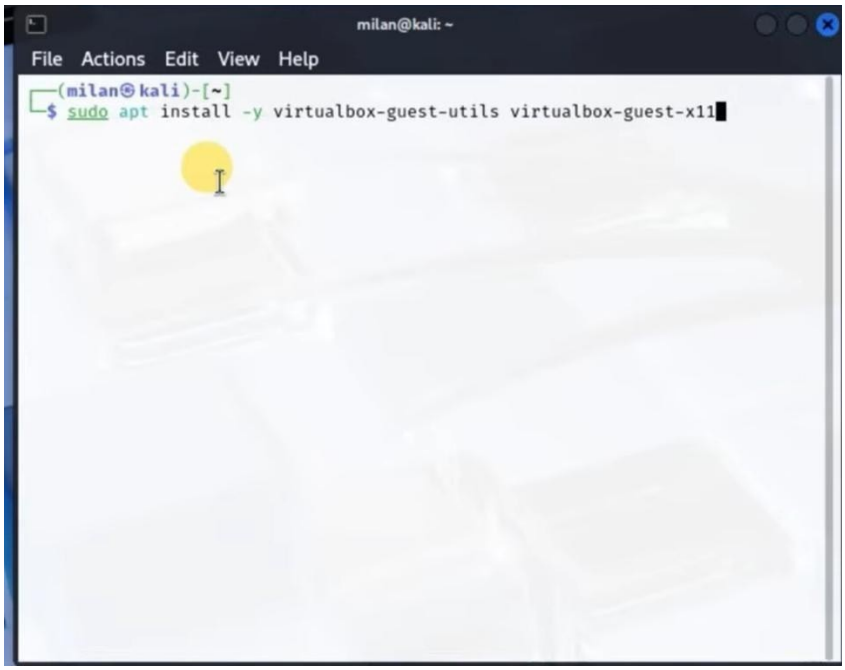
Open the Kali terminal and update the system with the following command:

```
sudo apt update && sudo apt upgrade -y
```



Next, install the VirtualBox guest additions so that screen resizing and clipboard sharing work:

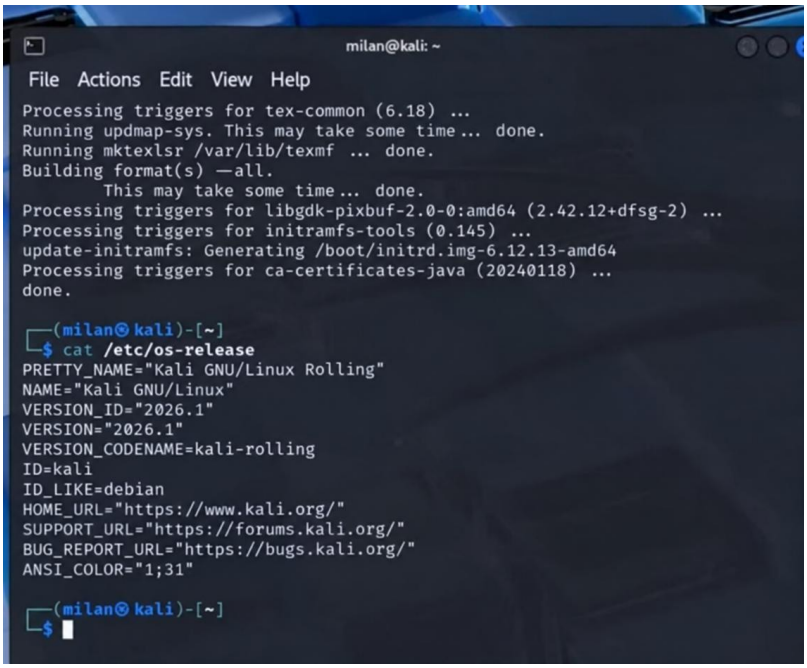
```
sudo apt install -y virtualbox-guest-utils virtualbox-guest-x11
```



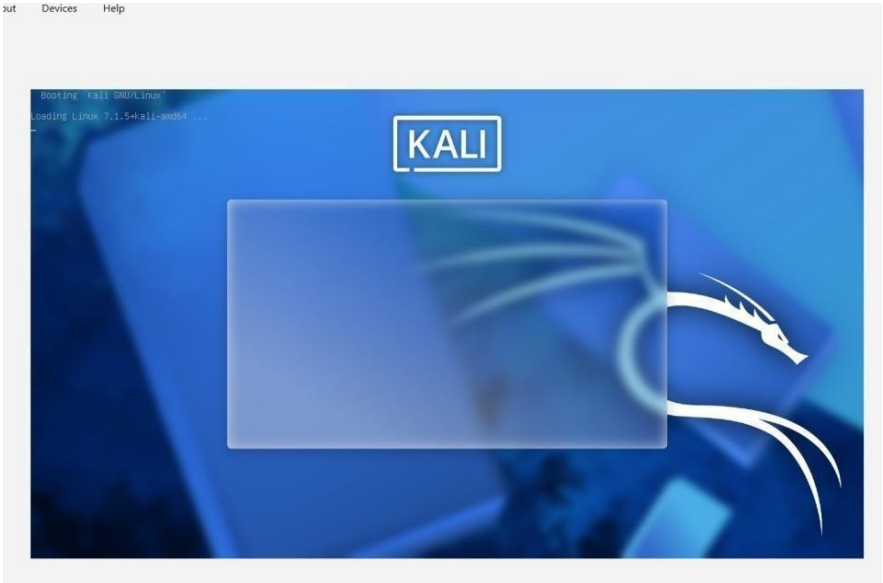
Step 6: Verify the Installation

Check the operating system details with the command below. The output shows Kali GNU/Linux Rolling, version 2026.1, confirming that Kali is installed correctly.

```
cat /etc/os-release
```



Restart the virtual machine to apply the changes and log in again to reach the Kali desktop.



Final Login

After the restart, the Kali login screen appears again. Sign in with the same details:

Username	Enter the credentials created during Kali Linux installation.
Password	Enter the credentials created during Kali Linux installation.

Once you are logged in, the cybersecurity lab is ready to use.

Conclusion

Kali Linux now runs as a virtual machine on VirtualBox in Windows 11, with 4 GB of memory and 2 CPUs. It was updated, fitted with guest additions and checked through the os-release output. The lab stays separate from the host system, so it is ready for the practical modules that follow.